

RESOLUCIÓN EJERCICIOS TP9

01)

$$a) \quad y \geq 2x^2 + 1$$

$$x + y \leq 4$$

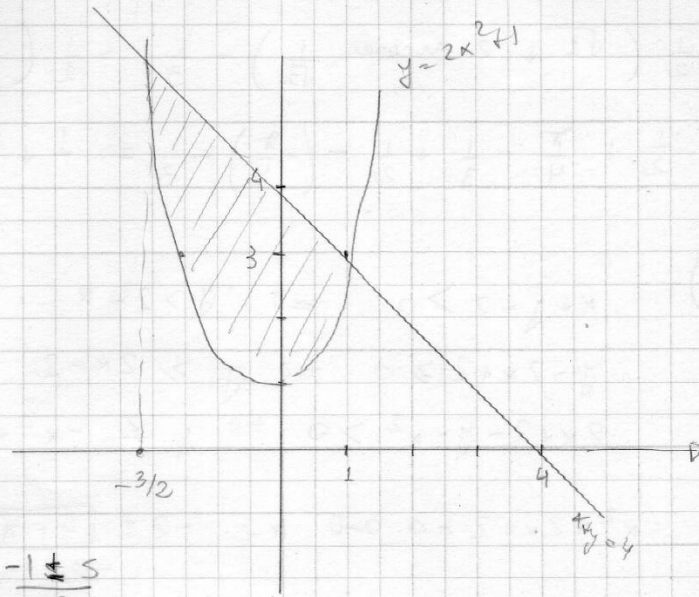
$$\left. \begin{array}{l} y = 2x^2 + 1 \\ y = 4 - x \end{array} \right\} = 0$$

$$\Rightarrow 2x^2 + 1 = 4 - x \Rightarrow$$

$$\Rightarrow 2x^2 + x - 3 = 0 \Rightarrow$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1 + 24}}{4} = \frac{-1 \pm 5}{4}$$

$$\Rightarrow x_1 = -\frac{3}{2} \wedge x_2 = 1 \Rightarrow x_1 = -\frac{3}{2} \wedge x_2 = 1$$



$$\text{Area} = \iint_D dx dy = \int_{-3/2}^1 dx \int_{2x^2+1}^{4-x} dy = \int_{-3/2}^1 dx (4 - x - 2x^2 - 1) =$$

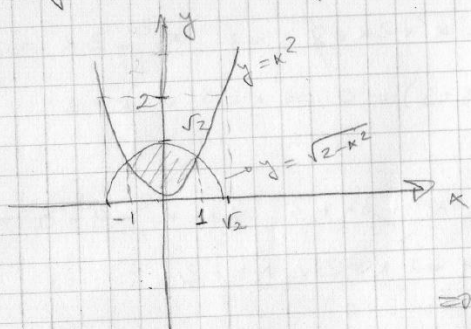
$$= \left[-\frac{x^2}{2} - \frac{2x^3}{3} + 3x \right]_{-3/2}^1 = -\frac{1}{2} - \frac{2}{3} + 3 + \frac{9/4}{2} + \frac{2(-27/8)}{3}$$

$$- 3\left(-\frac{3}{2}\right) = -\frac{1}{2} - \frac{2}{3} + 3 + \frac{9}{8} - \frac{54}{24} + \frac{9}{2} = \frac{125}{24}$$

1) b)

$$x^2 \leq y \leq \sqrt{2-x^2}$$

$$y = \sqrt{2-x^2} \Rightarrow y^2 = 2-x^2 \Rightarrow x^2 + y^2 = 2$$



$$\left. \begin{array}{l} x^2 = y \\ \sqrt{2-x^2} = y \end{array} \right\} \Rightarrow x^2 = \sqrt{2-x^2} \Rightarrow$$

$$\Rightarrow x^4 = 2-x^2 \Rightarrow$$

$$\Rightarrow x^4 + x^2 - 2 = 0 \Rightarrow$$

$$\Rightarrow x^2 = \frac{-1 \pm \sqrt{1+8}}{2} \quad \begin{array}{l} -2 \text{ absurdo} \\ 1 \end{array}$$

$$\Rightarrow x^2 = 1 \Rightarrow x = 1 \vee x = -1$$

$$\text{Area} = \int_{-1}^1 dx \int_{x^2}^{\sqrt{2-x^2}} dy = \int_{-1}^1 \left(\sqrt{2-x^2} - x^2 \right) dx =$$

$$= \left[\frac{1}{2} \left(x \sqrt{2-x^2} + 2 \arccos \frac{x}{\sqrt{2}} \right) - \frac{x^3}{3} \right]_{-1}^1 =$$

$$= \frac{1}{2} \left(\sqrt{1} + 2 \arccos \frac{1}{\sqrt{2}} \right) - \frac{1}{3} - \frac{1}{2} \left(-1 \sqrt{1} + 2 \arccos \left(\frac{-1}{\sqrt{2}} \right) \right) + \frac{(-1)}{3} =$$

$$= \frac{1}{2} + \frac{\pi}{4} - \frac{1}{3} + \frac{1}{2} - \left(-\frac{\pi}{4} \right) - \frac{1}{3} = 1 + \frac{\pi}{2} - \frac{2}{3} = \frac{\pi}{2} + \frac{1}{3}$$

$$1c) x+y-2 > 0 \Rightarrow y > 2-x$$

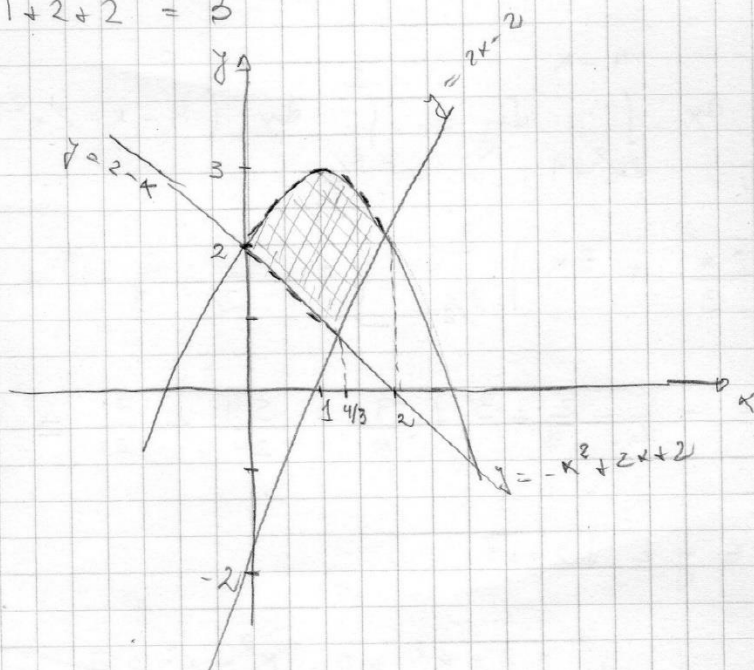
$$y-2x+2 > 0 \Rightarrow y > 2x-2$$

$$2x+2-y-x^2 > 0 \Rightarrow y < -x^2+2x+2$$

$$-x^2+2x+2=0 \Rightarrow x = \frac{-2 \pm \sqrt{4+8}}{-2} = \frac{-2 \pm \sqrt{12}}{-2} = 1 \pm \sqrt{3}$$

$$x_V = \frac{1+\sqrt{3} + 1-\sqrt{3}}{2} = 1$$

$$y_V = -1+2+2 = 3$$



$$\left. \begin{array}{l} y = 2-x \\ y = -x^2+2x+2 \end{array} \right\} \Rightarrow \cancel{2-x} = -x^2 + \cancel{2x} + \cancel{2} \Rightarrow x^2 - 3x = 0 \Rightarrow x(x-3) = 0 \Rightarrow x=0 \vee x=3$$

$$\left. \begin{array}{l} y = 2x-2 \\ y = -x^2+2x+2 \end{array} \right\} \Rightarrow \cancel{2x}-2 = -x^2 + \cancel{2x} + 2 \Rightarrow x^2 - 4 = 0 \Rightarrow x^2 = 4 \Rightarrow x=2 \vee x=-2$$

$$\left. \begin{array}{l} y = 2x - 2 \\ y = 2 - x \end{array} \right\} \Rightarrow 2x - 2 = 2 - x \Rightarrow 3x - 4 = 0 \Rightarrow x = \frac{4}{3}$$

$$\text{Area} = \int_0^{4/3} dx \int_{2-x}^{-x^2+2x+2} dy + \int_{4/3}^2 dx \int_{2x-2}^{-x^2+2x+2} dy =$$

$$= \int_0^{4/3} dx (-x^2 + 2x + 2 - (-2 + x)) + \int_{4/3}^2 dx (-x^2 + 2x + 2 - (2x - 2)) =$$

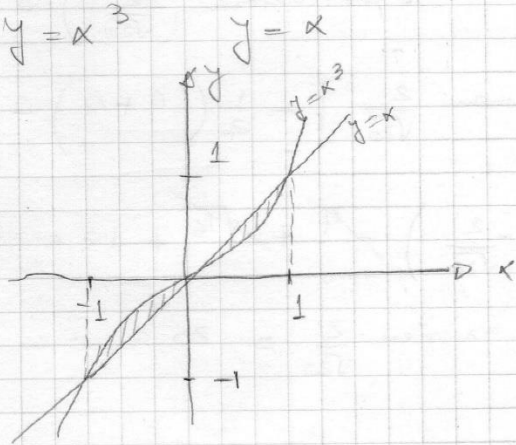
$$= \int_0^{4/3} \left(-\frac{x^3}{3} + \frac{3x^2}{2} \right) \Big|_0^{4/3} + \left(-\frac{x^3}{3} + 4x \right) \Big|_{4/3}^2 =$$

$$= -\frac{1}{3} \frac{4^3}{3^3} + \frac{3}{2} \frac{16}{9} + \left(-\frac{8}{3} \right) + 8 + \left(\frac{4}{3} \right)^3 \frac{1}{3} - 4 \cdot \frac{4}{3} =$$

$$= -\frac{64}{81} + \frac{48}{18} - \frac{8}{3} + 8 + \frac{64}{81} - \frac{16}{3} = \frac{8}{3}$$

1 2)

$$y = x^3$$



$$\begin{cases} y = x \\ y = x^3 \end{cases} = 0 \quad x = x^3 \Rightarrow x^3 - x = 0 \Rightarrow x(x^2 - 1) = 0 = 0$$

$$\Rightarrow x = 0 \vee x = 1 \vee x = -1$$

$$\text{Area } D = \iint_D 1 \, dx \, dy$$

$$A = \int_{-1}^0 dx \int_x^{x^3} dy + \int_0^1 dx \int_{x^3}^x dy = \int_{-1}^0 dx (x^3 - x) +$$

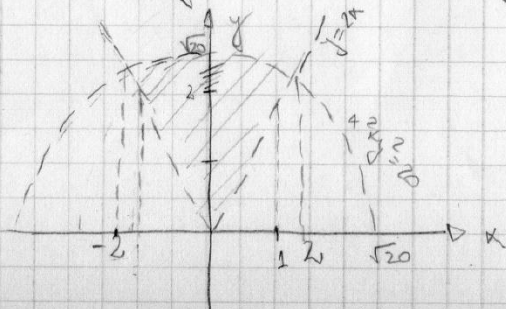
$$+ \int_0^1 dx (x - x^3) = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 =$$

$$= 0 - \frac{(-1)^4}{4} + \frac{(-1)^2}{2} + \frac{1}{2} - \frac{1}{4} = -\frac{1}{4} + \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{1}{2}$$

e) positiv $\rightarrow f(x, y) = (y - 2|x|) \sqrt{20 - x^2 - y^2}$
 ≥ 0

i) $y - 2|x| > 0 \Rightarrow y > 2|x|$

ii) $20 - x^2 - y^2 > 0 \Rightarrow x^2 + y^2 < 20$



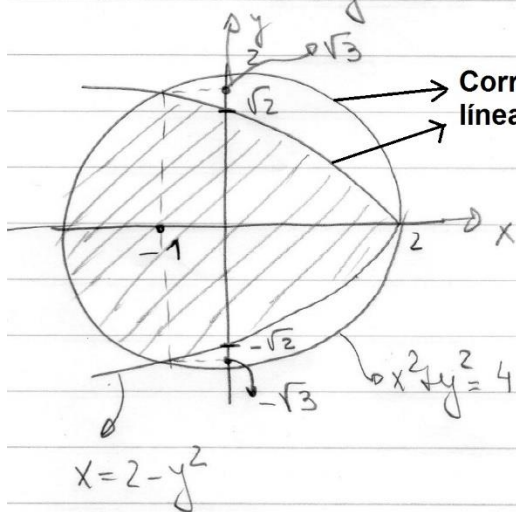
For $y = 2x$
 $x^2 + y^2 = 20 \Rightarrow x^2 + 4x^2 = 20 \Rightarrow 5x^2 = 20 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$
 $y = 2x \Rightarrow y = \pm 4$

Si $y = -2x \rightarrow y = -4$ at $x = 2$

$$\begin{aligned} \text{Area} &= \int_{-2}^0 dx \int_{-2x}^{\sqrt{20-x^2}} dy + \int_0^2 dx \int_{2x}^{\sqrt{20-x^2}} dy = \\ &= \int_{-2}^0 dx (\sqrt{20-x^2} + 2x) + \int_0^2 dx (\sqrt{20-x^2} - 2x) = \\ &= \left(\frac{1}{2} \left[x \sqrt{20-x^2} + 20 \arcsin \frac{x}{\sqrt{20}} \right] \Big|_{-2}^0 \right) + \left(x^2 \Big|_{-2}^0 \right) + \\ &+ \left(\frac{1}{2} \left[x \sqrt{20-x^2} + 20 \arcsin \frac{x}{\sqrt{20}} \right] \Big|_0^2 \right) - \left(x^2 \Big|_0^2 \right) = \\ &= \frac{1}{2} (0 + 0) - \frac{1}{2} \left(-2 \sqrt{20-4} + 20 \arcsin \left(\frac{-2}{\sqrt{20}} \right) \right) + 0 - (-2)^2 + \\ &+ \frac{1}{2} \left[2 \sqrt{20-4} + 20 \arcsin \frac{2}{\sqrt{20}} \right] - \frac{1}{2} (0 + 0) - 4 = \\ &= \cancel{\sqrt{16}} + 10 \arcsin \left(\frac{-2}{\sqrt{20}} \right) + \cancel{\sqrt{16}} + 10 \arcsin \frac{2}{\sqrt{20}} - 4 = \\ &= 10 \arcsin \frac{2}{\sqrt{20}} + 10 \arcsin \frac{2}{\sqrt{20}} = 20 \arcsin \frac{2}{\sqrt{20}} \end{aligned}$$

$$1) f) \quad 4 - x^2 - y^2 > 0 \wedge 2 - x - y^2 > 0$$

$$x^2 + y^2 < 4 \wedge x < 2 - y^2$$



$$\left. \begin{array}{l} 2 - y^2 = x \\ x^2 + y^2 = 4 \end{array} \right\} = 0$$

$$\Rightarrow x^2 + 2 - x = 4 \Rightarrow$$

$$\Rightarrow x^2 - x - 2 = 0 \Rightarrow x_1 = 2$$

$$\Rightarrow x_2 = -1$$

$$2 - y^2 = 0 \Rightarrow |y| = \sqrt{2} \Rightarrow y_1 = \sqrt{2}$$

$$\Rightarrow y_2 = -\sqrt{2}$$

$$x = -1 \Rightarrow -1 = 2 - y^2 \Rightarrow y^2 = 3 \Rightarrow y_1 = \sqrt{3}$$

$$\Rightarrow y_2 = -\sqrt{3}$$

$$Area = \int_{-\sqrt{3}}^{\sqrt{3}} dy \int_{-\sqrt{4-y^2}}^{2-y^2} dx = \int_{-\sqrt{3}}^{\sqrt{3}} (2 - y^2 + \sqrt{4-y^2}) dy =$$

$$= \left[2y - \frac{y^3}{3} + \frac{1}{2} \left(y - \sqrt{4-y^2} + 4 \arccos\left(\frac{y}{2}\right) \right) \right]_{-\sqrt{3}}^{\sqrt{3}} =$$

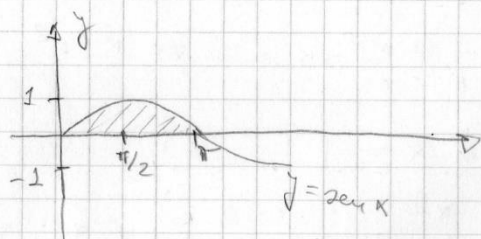
$$= 2\sqrt{3} - \frac{3\sqrt{3}}{3} + \frac{1}{2} \left(\sqrt{3} - \sqrt{1} + 4 \arccos\left(\frac{\sqrt{3}}{2}\right) \right) - 2(-\sqrt{3}) +$$

$$+ \frac{3(-\sqrt{3})}{3} - \frac{1}{2} \left(-\sqrt{3} - \sqrt{1} + 4 \arccos\left(-\frac{\sqrt{3}}{2}\right) \right) =$$

$$= 2\sqrt{3} - \sqrt{3} + \frac{\sqrt{3}}{2} - \frac{1}{2} + 2 \cdot \frac{\pi}{3} + 2\sqrt{3} - \sqrt{3} + \frac{\sqrt{3}}{2} + \frac{1}{2} -$$

$$- 2\left(-\frac{\pi}{3}\right) = 3\sqrt{3} + \frac{4\pi}{3}$$

2a)



$$\iint_D dx dy = \int_0^{\pi} dx \int_0^{\sin x} dy = \int_0^{\pi} \sin x dx = \left[-\cos x \right]_0^{\pi} =$$

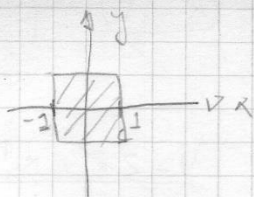
$$= -\cos \pi + \cos 0 = -(-1) + 1 = 2$$

$$= \int_0^1 dy \int_{\arcsin y}^{\pi - \arcsin y} dx = \int_0^1 dy (\pi - \arcsin y - \arcsin y) =$$

$$= \left[\pi y \right]_0^1 - 2 \left[y \arcsin y + \sqrt{1-y^2} \right]_0^1 = \pi - 2$$

$$\left(\frac{\pi}{2} + \sqrt{0} - 0 - 1 \right) = \pi - \pi + 2 = 2$$

2b)



$$\iint_D x dx dy = \int_{-1}^1 dx \int_{-1}^1 x dy =$$

$$= \int_{-1}^1 x dx (1+1) = 2 \left[\frac{x^2}{2} \right]_{-1}^1 =$$

$$= 2 \left[\frac{1}{2} - \frac{1}{2} \right] = 0$$

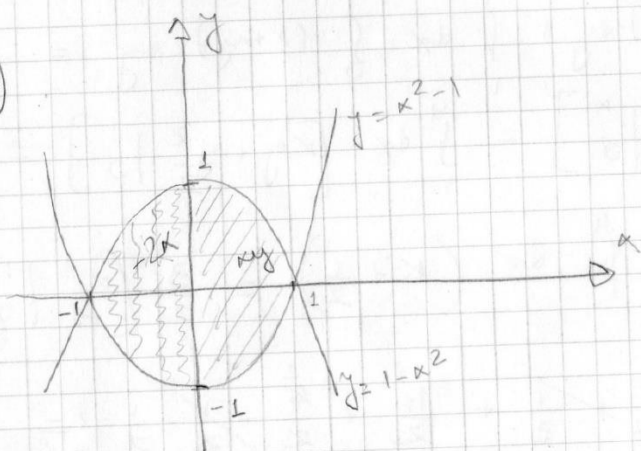
$$\iint_D x dx dy = \int_{-1}^1 dy \int_{-1}^1 x dx = \int_{-1}^1 dy \left[\frac{x^2}{2} \right]_{-1}^1 =$$

$$= \int_{-1}^1 dy \left(\frac{1}{2} - \frac{1}{2} \right) = 0$$

$$\begin{aligned}
 2) c) \quad \iint_D |x| \, dx \, dy &= \int_{-1}^1 dx \int_{-1}^1 |x| \, dy = \int_{-1}^1 |x| \, dx (1+1) = \\
 &= 2 \left[\int_{-1}^0 (-x) \, dx + \int_0^1 x \, dx \right] = \\
 &= 2 \left[-\frac{x^2}{2} \Big|_{-1}^0 \right] + 2 \left[\frac{x^2}{2} \Big|_0^1 \right] = \\
 &= 2 \left(-0 + \frac{1}{2} \right) + 2 \left(\frac{1}{2} - 0 \right) = 1+1=2
 \end{aligned}$$

$$\begin{aligned}
 \iint_D |x| \, dx \, dy &= \int_{-1}^1 dy \int_{-1}^1 |x| \, dx = \int_{-1}^1 dy \left(\int_{-1}^0 -x \, dx + \right. \\
 &+ \left. \int_0^1 x \, dx \right) = \int_{-1}^1 dy \left(-\frac{x^2}{2} \Big|_{-1}^0 + \frac{x^2}{2} \Big|_0^1 \right) = \\
 &= \int_{-1}^1 dy \left(-0 + \frac{1}{2} + \frac{1}{2} - 0 \right) = \int_{-1}^1 dy = 1+1=2
 \end{aligned}$$

2) d)



$$y = x^2 - 1 \Rightarrow x^2 = y + 1 \Rightarrow |x| = \sqrt{y+1}$$

$$y = 1 - x^2 \Rightarrow x^2 = 1 - y \Rightarrow |x| = \sqrt{1-y}$$

$$\begin{aligned}
 \iint_D f(x,y) \, dx \, dy &= \int_{-1}^0 dx \int_{x^2-1}^{1-x^2} -2x \, dy + \int_0^1 dx \int_{x^2-1}^{1-x^2} xy \, dy = \\
 &= \int_{-1}^0 -2x \, dx (1-x^2-x^2+1) + \int_0^1 x \, dx \left[\frac{y^2}{2} \Big|_{x^2-1}^{1-x^2} \right] =
 \end{aligned}$$

$$= \int_{-1}^0 (-4x + 4x^3) dx + \int_0^1 x dx \left[\frac{(-x^2)^2}{2} - \frac{(x^2-1)^2}{2} \right] =$$

$$= \left[-2x^2 + x^4 \right]_{-1}^0 + \int_0^1 dx \cdot x \cdot \frac{1 - 2x^2 + x^4 - x^4 - 1 + 2x^2}{2} =$$

$$= 0 + 2(-1)^2 - (-1)^4 + 0 = 2 - 1 = 1$$

$$\iint_D f(x,y) dx dy = \int_{-1}^0 dy \int_{-\sqrt{y+1}}^0 dx (-2x) + \int_0^1 dy \int_{-\sqrt{1-y}}^0 dx (-2x) +$$

$$+ \int_{-1}^0 dy \int_0^{\sqrt{y+1}} dx xy + \int_0^1 dy \int_0^{\sqrt{1-y}} dx xy =$$

$$= \int_{-1}^0 dy \left[-x^2 \right]_{-\sqrt{y+1}}^0 + \int_0^1 dy \left[-x^2 \right]_{-\sqrt{1-y}}^0 +$$

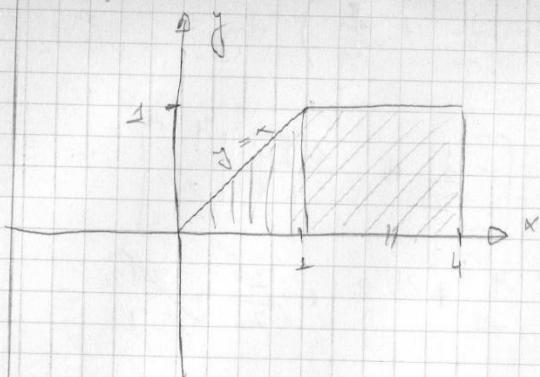
$$+ \int_{-1}^0 y dy \left[\frac{x^2}{2} \right]_0^{\sqrt{y+1}} + \int_0^1 y dy \left[\frac{x^2}{2} \right]_0^{\sqrt{1-y}} =$$

$$= \int_{-1}^0 (-0 + y+1) dy + \int_0^1 [-0 + 1-y] dy + \int_{-1}^0 y \frac{(y+1)}{2} dy +$$

$$+ \int_0^1 y \frac{(1-y)}{2} dy = \left[\frac{y^2}{2} + y \right]_{-1}^0 + \left[y - \frac{y^2}{2} \right]_0^1 + \frac{1}{2} \left[\frac{y^3}{3} + \frac{y^2}{2} \right]_{-1}^0 +$$

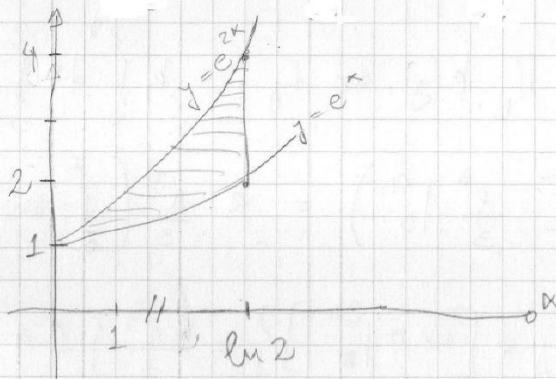
$$+ \frac{1}{2} \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1 = 0 - \frac{1}{2} + 1 - \frac{1}{2} + \frac{1}{2} \left(\frac{1}{3} - \frac{1}{2} \right) + \frac{1}{2} \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{2}$$

$$\begin{aligned}
 2e) \quad & \int_0^1 dy \int_0^x (x+xy) dy + \int_1^4 dx \int_0^1 (x+xy) dy = \\
 & = \int_0^1 dy \left[xy + \frac{y^2}{2} \Big|_0^x \right] = \int_0^1 dy \left[xy + \frac{y^2}{2} \Big|_0^1 \right] = \\
 & = \int_0^1 \left(x^2 + \frac{x^2}{2} \right) dx + \int_1^4 dx \left(x + \frac{1}{2} \right) = \frac{3}{2} \left[\frac{x^3}{3} \Big|_0^1 \right] + \\
 & + \left[\frac{x^2}{2} + \frac{x}{2} \Big|_1^4 \right] = \cancel{\frac{3}{2}} \cdot \frac{1}{3} + \frac{16}{2} + \frac{2}{2} - \cancel{\frac{1}{2}} - \frac{1}{2} = \\
 & = 10 - \frac{1}{2} = \frac{19}{2}
 \end{aligned}$$



$$\begin{aligned}
 & \int_0^1 dy \int_y^4 dx (x+xy) = \int_0^1 dy \left[\frac{x^2}{2} + yx \Big|_y^4 \right] = \\
 & = \int_0^1 \left(\frac{16}{2} + 4y - \frac{y^2}{2} - y^2 \right) dy = \left[8y + 2y^2 - \frac{3}{2} \frac{y^3}{3} \Big|_0^1 \right] \\
 & = 8 + 2 - \frac{1}{2} = 10 - \frac{1}{2} = \frac{19}{2}
 \end{aligned}$$

f)



$$e^{\ln 2} = 2$$

$$e^{2 \ln 2} = (e^{\ln 2})^2 = 4$$

$$y = e^{2x} \Rightarrow \ln y = 2x$$

$$y = e^x \Rightarrow \ln y = x$$

$$\iint_D e^{-x} dx dy = \int_0^{\ln 2} dx \int_{e^x}^{e^{2x}} e^{-x} dy =$$

$$= \int_0^{\ln 2} e^{-x} (e^{2x} - e^x) dx = \int_0^{\ln 2} (e^x - 1) dx =$$

$$= [e^x - x]_0^{\ln 2} = 2 - \ln 2 - 1 + 0 =$$

$$= 1 - \ln 2$$

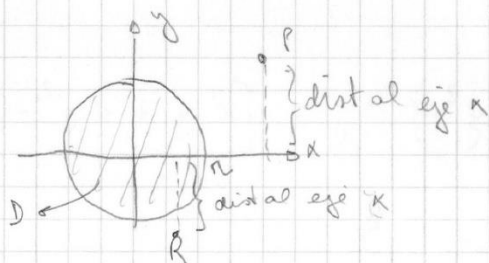
$$\int_1^2 dy \int_{\frac{\ln y}{2}}^{\ln y} e^{-x} dx + \int_2^4 dy \int_{\frac{\ln y}{2}}^{\ln 2} e^{-x} dx =$$

$$= \int_1^2 dy \left[-e^{-x} \right]_{\frac{\ln y}{2}}^{\ln y} + \int_2^4 dy \left[-e^{-x} \right]_{\frac{\ln y}{2}}^{\ln 2} =$$

$$= \int_1^2 dy \left[-e^{-\ln y} + e^{-\frac{\ln y}{2}} \right] + \int_2^4 dy \left[-e^{-\ln 2} + e^{-\frac{\ln y}{2}} \right] =$$

$$\begin{aligned}
 &= \int_1^2 \left(-\frac{1}{y} + \frac{1}{\sqrt{y}} \right) dy + \int_2^4 \left(2 - \frac{1}{2} + \frac{1}{\sqrt{y}} \right) dy = \\
 &= \int_1^2 \left[-\ln|y| + 2\sqrt{y} \right]_1^2 + \left[-\frac{y}{2} + \frac{2\sqrt{y}}{\frac{3}{2}} \right]_2^4 = \\
 &= -\ln 2 + 2\sqrt{2} + 0 - 2 - \frac{4}{2} + 2 \cdot 2 + \frac{2}{2} - \frac{2\sqrt{2}}{\frac{3}{2}} = \\
 &= -\ln 2 - \cancel{2} - \cancel{2} + 4 + 1 = 1 - \ln 2
 \end{aligned}$$

03)



$$S = k|y| \quad (k \in \mathbb{R})$$

$$\begin{aligned}
 m &= \iint_D S \, dx \, dy = \iint_D k|y| \, dx \, dy = k \int_0^{2\pi} d\theta \int_0^R \rho \, \rho |\sin \theta| \, d\rho = \\
 &= k \int_0^{2\pi} |\sin \theta| \, d\theta \left[\frac{\rho^3}{3} \right]_0^R = k \frac{R^3}{3} \left[\int_0^{\pi} \sin \theta \, d\theta + \int_{\pi}^{2\pi} (-\sin \theta) \, d\theta \right] \\
 &= k \frac{R^3}{3} \left\{ \left[-\cos \theta \right]_0^{\pi} + \left[\cos \theta \right]_{\pi}^{2\pi} \right\} = \frac{k R^3}{3} [1 + 1 + 1 + 1] = \\
 &= \frac{4}{3} k R^3
 \end{aligned}$$

$$x_G = \frac{\iint_D k \, S \, dx \, dy}{\iint_D S \, dx \, dy} = \frac{\iint_D k \times |y| \, dx \, dy}{\frac{4}{3} k R^3} =$$

$$= \frac{\cancel{k} \int_0^{2\pi} d\theta \int_0^R \rho^2 \cos\theta \sin\theta | \rho \, d\rho}{4/3 \cancel{k} R^3} = \frac{3}{4} \frac{\int_0^{2\pi} \cos\theta \sin\theta \left[\frac{\rho^4}{4} \Big|_0^R \right] d\theta}{R^3} =$$

$$= \frac{3}{4 R^3} \cdot \frac{R^4}{4} \left[\int_0^{\pi} \cos\theta \sin\theta \, d\theta + \int_{\pi}^{2\pi} -\sin\theta \cos\theta \, d\theta \right] =$$

$$= \frac{3}{16} R \left[\frac{\sin^2\theta}{2} \Big|_0^{\pi} - \frac{\sin^2\theta}{2} \Big|_{\pi}^{2\pi} \right] =$$

$$= \frac{3}{16} R [0 - 0 - 0 + 0] = 0$$

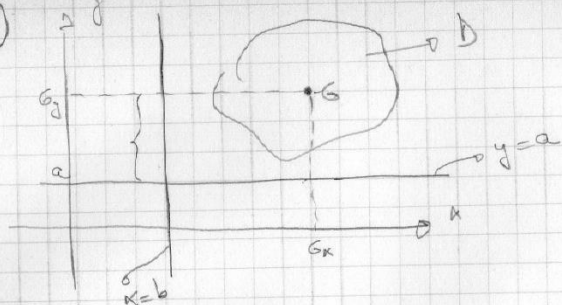
$$y_G = \frac{\iint_D y \, dxdy}{4/3 k R^3} = \frac{\iint_D y \cdot k |y| \, dxdy}{4/3 k R^3} = \frac{\cancel{k} \int_0^{2\pi} d\theta \int_0^R \sin\theta |\sin\theta| \rho^2 \, d\rho}{4/3 \cancel{k} R^3}$$

$$= \frac{3}{4 R^3} \int_0^{2\pi} \sin\theta |\sin\theta| \left[\frac{\rho^4}{4} \Big|_0^R \right] d\theta = \frac{3 R^4}{4 R^3 4}$$

$$\left[\int_0^{\pi} \sin^2\theta \, d\theta + \int_0^{2\pi} -\sin^2\theta \, d\theta \right] = 0$$

$$G = (x_G, y_G) = (0, 0)$$

04)



$$i) I_a = \iint_D |G_y - a|^2 \delta \, dx \, dy = \iint_D (G_y - a)^2 \delta \, dx \, dy =$$

$$\iint_D (G_y^2 + a^2 - 2G_y a) \delta \, dx \, dy = \iint_D G_y^2 \delta \, dx \, dy +$$

$$+ \iint_D a^2 \delta \, dx \, dy - 2 \iint_D G_y a \delta \, dx \, dy \geq 0 \quad (\text{por ser}$$

un momento de inercia)

Si la recta pasa por $G \rightarrow$ la ecuación es: $y = G_y \quad (a = G_y)$

Luego la expresión se reduce a:

$$\cancel{\iint_D G_y^2 \delta \, dx \, dy} + \cancel{\iint_D G_y^2 \delta \, dx \, dy} - 2 \iint_D G_y G_y \delta \, dx \, dy =$$

$= 0 \rightarrow$ donde 0 es el mínimo valor posible que toma el momento de inercia.

Por lo tanto, si la recta pasa por $G \rightarrow y = G_y$ el I_{Gy} es mínimo y vale 0

$$ii) I_b = \iint_D |G_x - b|^2 \, d x \, d y = \iint_D (G_x - b)^2 \, d x \, d y \geq 0 \text{ (por ser un momento de inercia)}$$

Si la recta pasa por $G \rightarrow$ su ecuación es: $x = G_x$ ($b = G_x$)

Luego la expresión se reduce a:

$$\iint_D (G_x - G_x)^2 \, d x \, d y = 0 \rightarrow \text{donde 0 es el mínimo valor posible que toma el momento de inercia}$$

5)

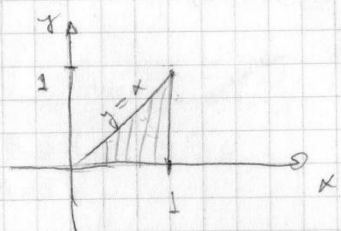
$$a) \int_0^1 \int_0^{\sqrt{1-x^2}} x \, d y \, d x = \int_0^1 x \, d x \int_0^{\sqrt{1-x^2}} d y = \int_0^1 x \sqrt{1-x^2} \, d x =$$

$$= \left[-\frac{1}{3} \sqrt{(1-x^2)^3} \right]_0^1 = -0 + \frac{1}{3} \sqrt{1} = \frac{1}{3}$$

$$5b) \int_0^1 \int_0^1 e^{x^2} dy dx = \int_0^1 dy \int_0^1 e^{x^2} dx = ?$$

no tiene primitiva

Veo aplicando un cambio en el orden de integración:



$$\int_0^1 dx \int_0^x e^{x^2} dy = \int_0^1 dx x e^{x^2} = (A)$$

$$u = e^{x^2}$$

$$du = \frac{e^{x^2}}{u} \cdot 2x dx$$

$$x=0 \Rightarrow u = e^{0^2} = 1$$

$$x=1 \Rightarrow u = e^{1^2} = e$$

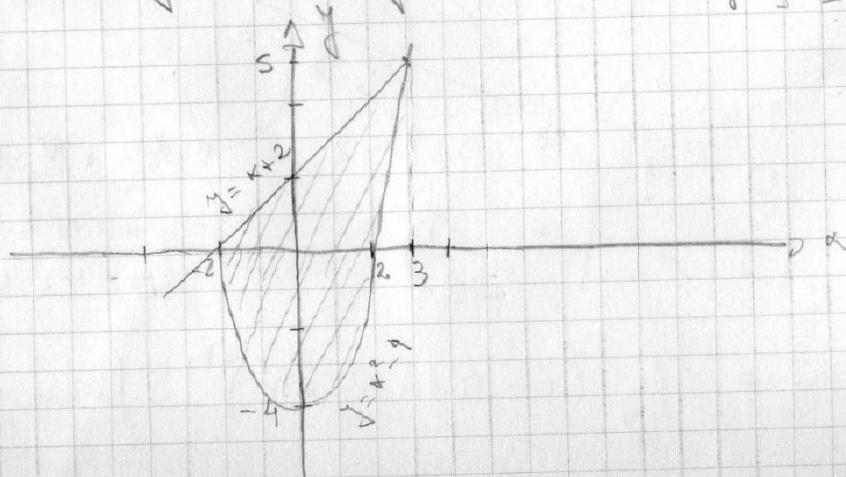
$$(A) = \int_1^e \frac{du}{u 2x} \cdot x = \frac{1}{2} \left[\ln u \right]_1^e = \frac{1}{2} (e - 1)$$

$$5c) \int_{-4}^0 dy \int_{-\sqrt{y+4}}^{\sqrt{y+4}} dx + \int_0^5 dy \int_{y-2}^{\sqrt{y+4}} dx$$

Ves cambiando el orden de integración

$$x = \sqrt{y+4} \Rightarrow x^2 - 4 = y \quad \begin{cases} y=5 \Rightarrow x=3 \\ y=-4 \Rightarrow x=0 \\ y=0 \Rightarrow x=\pm 2 \end{cases}$$

$$x = y - 2 \Rightarrow y = x + 2 \quad \begin{cases} y=5 \Rightarrow x=3 \end{cases}$$



$$\begin{aligned} \int_{-2}^3 dx \int_{x^2-4}^{x+2} dy &= \int_{-2}^3 dx (x+2 - x^2 + 4) = \left[-\frac{x^3}{3} + \frac{x^2}{2} + 6x \right]_{-2}^3 = \\ &= -\frac{27}{3} + \frac{9}{2} + 18 + \frac{8}{3} - \frac{4}{2} - (-12) = -9 + \frac{9}{2} + 18 - \\ &- \frac{8}{3} - 2 + 12 = \frac{9}{2} - \frac{8}{3} + 19 = \frac{27-16}{6} + 19 = \frac{11}{6} + 19 \end{aligned}$$

$$e) a) \iint_D (6-x-y)^{-1} dx dy$$

$$D: |x+y| \leq 2 \wedge y \leq x+2 \leq 4 \rightarrow \begin{cases} x=r \\ y=u-r \end{cases}$$

usando $(x,y) = (r, u-r)$

Hallamos el recinto transformado de D

$$|x+y| \leq 2 \Rightarrow -2 \leq x+y \leq 2$$

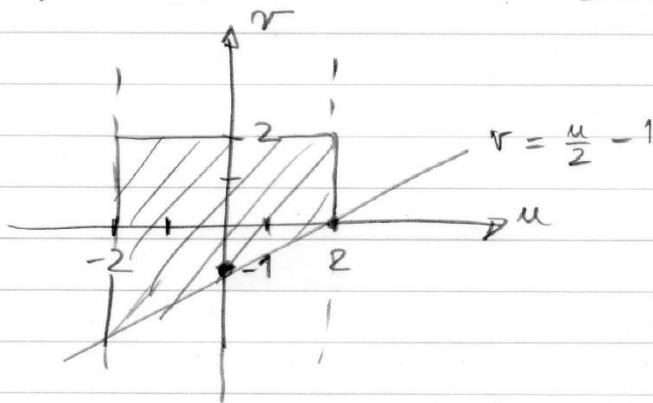
$$i) -2 \leq x+y \rightarrow -2 \leq r + u - r = u \Rightarrow \boxed{u \geq -2}$$

$$ii) x+y \leq 2 \rightarrow r + u - r \leq 2 \Rightarrow \boxed{u \leq 2}$$

$$y \leq x+2 \leq 4$$

$$i) y \leq x+2 \rightarrow u-r \leq r+2 \Rightarrow 2r \geq u-2 \Rightarrow \boxed{r \geq \frac{u}{2} - 1}$$

$$ii) x+2 \leq 4 \Rightarrow x \leq 2 \rightarrow \boxed{r \leq 2}$$



$$J = \frac{\partial(x,y)}{\partial(u,r)} = \begin{vmatrix} x'_u & x'_r \\ y'_u & y'_r \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1$$

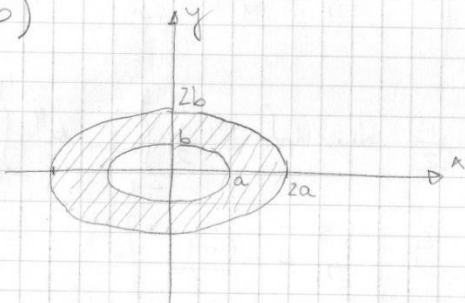
$$|J| = |-1| = 1$$

$$\iint_D (6-x-y)^{-1} dx dy = \int_{-2}^2 du \int_{\frac{u}{2}-1}^2 dr \frac{1}{6-r-u+r} \cdot 1$$

$$= \int_{-2}^2 \ln \frac{1}{6-u} \left(2 - \frac{u}{2} + 1 \right) du = \int_{-2}^2 \frac{3 - u/2}{6-u} du =$$

$$= \int_{-2}^2 \frac{1}{2} \frac{\cancel{6-u}}{6-u} du = \frac{1}{2} (2 - (-2)) = \boxed{2}$$

b)



$$\begin{cases} x = a \rho \cos \theta \\ y = b \rho \sin \theta \end{cases}$$

$$1 \leq \rho \leq 2$$

$$0 \leq \theta \leq 2\pi$$

$$J = \frac{\partial(x, y)}{\partial(\rho, \theta)} = \begin{vmatrix} a \cos \theta & -a \rho \sin \theta \\ b \sin \theta & b \rho \cos \theta \end{vmatrix} = ab \rho \cos^2 \theta + ab \rho \sin^2 \theta =$$

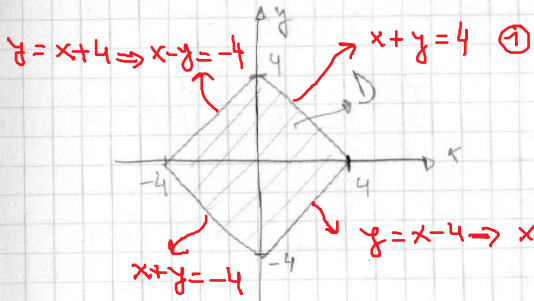
$$= ab \rho$$

$$\text{Area} = \iint_R dx dy = \iint_W ab \rho \, d\rho \, d\theta = ab \int_1^2 \rho \, d\rho \int_0^{2\pi} d\theta =$$

$$= ab \, 2\pi \left[\frac{\rho^2}{2} \right]_1^2 = 2ab\pi \left(\frac{4}{2} - \frac{1}{2} \right) = 3ab\pi$$

Como se trata de una transformación lineal, los segmentos se transforman en segmentos. Por ello podemos realizar los transformados de los extremos para evaluar su imagen en el plano (uv) , detallado en (*)

C) D: $|x| + |y| \leq 4$



$$\begin{cases} x - y = u \\ x + y = v \end{cases}$$

① $x+y=4 \sim 0 \leq x \leq 4$

$$r=4 \quad \wedge \quad 0 \leq \frac{n+r}{2} \leq 4$$

$$0 \leq n + n \leq 8$$

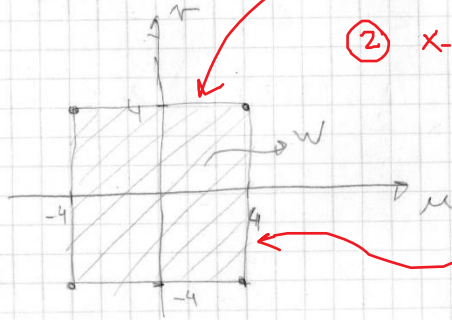
$$N=4 \wedge -4 \leq n \leq 4$$

② $x - y = 4 \quad \sim \quad 0 \leq x \leq 4$

$$n=4 \wedge 0 \leq \frac{n+5}{2} \leq 4$$

$$0 \leq \frac{1 + \sqrt{5}}{4} \leq 8$$

$$n=4 \wedge -4 \leq v \leq 4$$



$$\begin{aligned} (*) \quad T(4, 0) &= (4, 4) \\ T(0, 4) &= (-4, 4) \\ T(-4, 0) &= (-4, -4) \\ T(0, -4) &= (4, -4) \end{aligned}$$

$$\begin{cases} x - y = u \\ x + y = v \end{cases}$$

$$\Rightarrow \textcircled{+} \text{MAM} \rightarrow ZK = u + v \Rightarrow K = \frac{u+v}{2}$$

$$\textcircled{+} \text{MAM} \rightarrow -2y = u - r = 0 \quad y = \frac{u - r}{-2} = \frac{r - u}{2}$$

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$\begin{aligned} \iint_D (x-y)^4 \, dx \, dy &= \iint_W u^4 \cdot \frac{1}{2} \, du \, dv = \frac{1}{2} \int_{-4}^4 du u^4 \int_{-4}^4 dv = \\ &= \frac{1}{2} (4+4) \left[\frac{u^5}{5} \right]_{-4}^4 = 4 \left[\frac{4^5}{5} - \frac{(-4)^5}{5} \right] = \\ &= 4 \cdot 2 \cdot \frac{4^5}{5} = 2 \cdot \frac{4^6}{5} \end{aligned}$$

$$6)d) \iint_D (x+y-2)^2 dx dy \quad \begin{cases} x = u+v \\ y = u-v \end{cases}$$

$$D = \{(x,y) \in \mathbb{R}^2 / y \geq |x|, x+2y \leq 3\}$$

Hallamos el recinto transformado de D

$$i) |x| \leq y \rightarrow |u+v| \leq u-v \Rightarrow$$

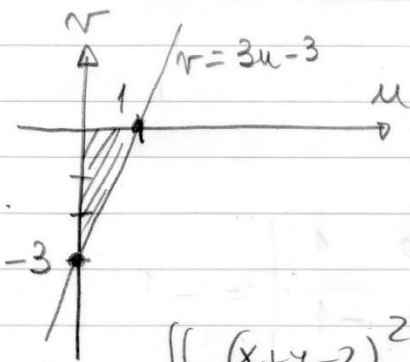
$$\Rightarrow \underbrace{-(u-v) \leq u+v}_{(1)} \leq \underbrace{u-v}_{(2)} \Rightarrow$$

$$(1): -u+v \leq u+v \Rightarrow 2u \geq 0 \Rightarrow \boxed{u \geq 0}$$

$$(2): u+v \leq u-v \Rightarrow 2v \leq 0 \Rightarrow \boxed{v \leq 0}$$

$$ii) x+2y \leq 3 \rightarrow u+v+2(u-v) \leq 3 \Rightarrow$$

$$\Rightarrow \boxed{3u-3 \leq v}$$



$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x'_u & x'_v \\ y'_u & y'_v \end{vmatrix} =$$

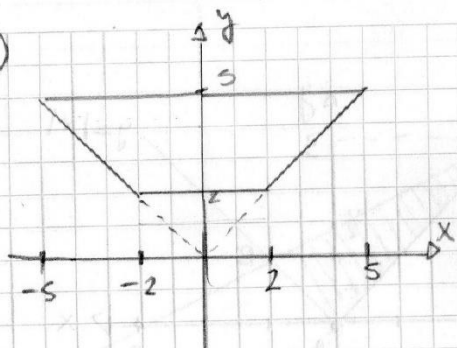
$$= \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2 \Rightarrow |J| = 2$$

$$\iint_D (x+y-2)^2 dx dy = \int_0^1 du \int_{3u-3}^0 dv (u+v+u-v-2)^2 \cdot 2 =$$

$$= 2 \int_0^1 du (2u-2)^2 \left[v \Big|_{3u-3}^0 \right] = 2 \int_0^1 du (2u-2)^2 (-3u+3) =$$

$$= 24 \int_0^1 (-u^3 + 3u^2 - 3u + 1) du = \left. -\frac{u^4}{4} + u^3 - \frac{3u^2}{2} + u \right|_0^1 = \boxed{6}$$

e)



$$\frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}$$

$$y=2 \rightarrow \rho \sin \theta = 2 \Rightarrow \rho = \frac{2}{\sin \theta}$$

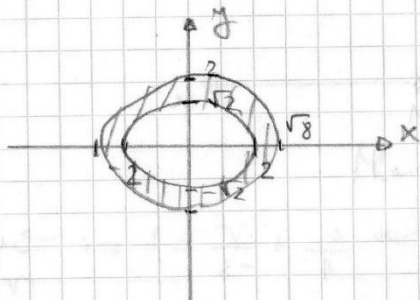
$$y=5 \rightarrow \rho \sin \theta = 5 \Rightarrow \rho = \frac{5}{\sin \theta}$$

$$\iint_D y (x^2 + y^2)^{-1} dx dy = \int_{\pi/4}^{3\pi/4} \int_{\frac{2}{\sin \theta}}^{\frac{5}{\sin \theta}} \frac{\rho \sin \theta}{\rho^2} \rho d\rho d\theta =$$

$$= \int_{\pi/4}^{3\pi/4} \sin \theta \left(\frac{5}{\sin \theta} - \frac{2}{\sin \theta} \right) d\theta = 3 \left(\frac{3\pi}{4} - \frac{\pi}{4} \right) =$$

$$= 3 \frac{2\pi}{4} = \frac{6\pi}{4} = \frac{3\pi}{2}$$

7) a) $f(x,y) = e^{x^2+y^2} = e^4 \Rightarrow x^2+y^2=4 \Rightarrow \frac{x^2}{4} + \frac{y^2}{2} = 1$
 $e^{x^2+y^2} = e^8 \Rightarrow x^2+y^2=8 \Rightarrow \frac{x^2}{8} + \frac{y^2}{4} = 1$



$$1 \leq \frac{x^2}{4} + \frac{y^2}{2} \leq 2$$

$$0 \leq \theta \leq 2\pi$$

$$1 \leq \rho \leq \sqrt{2}$$

Aplicando y 6) b)

$$\begin{aligned} x &= \rho \cos \theta \\ y &= \rho \sin \theta \\ J &= \rho \end{aligned}$$

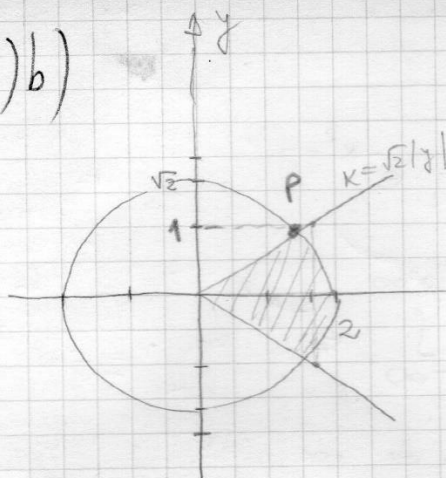
$$\begin{aligned} \text{Area } D &= \iint_D dx dy = \iint_D \rho d\rho d\theta = \int_0^{2\pi} \int_1^{\sqrt{2}} \rho d\rho d\theta = \\ &= 2\pi \left[\frac{\rho^2}{2} \right]_1^{\sqrt{2}} = 2\pi \left(\frac{2}{2} - \frac{1}{2} \right) = \frac{1}{2} 2\pi = \pi \end{aligned}$$

$$07) b) \iint_D e^{x^2+2y^2} dx dy \quad \text{con} \quad x^2+2y^2 \leq 4$$

$$x \geq \sqrt{2}|y|$$

$$x^2+2y^2=4 \Rightarrow \frac{x^2}{4} + \frac{y^2}{2} = 1$$

2) b)



$$x = \sqrt{2}|y| \Rightarrow x^2 = 2y^2$$

on the ellipse $\rightarrow 2y^2 + 2y^2 = 4 \Rightarrow |y| = 1$
 $\Rightarrow y = 1 \vee y = -1$

$$P = (\sqrt{2}, 1)$$

$$\begin{cases} \sqrt{2} = 2 \cdot 1 \cdot \cos \theta \Rightarrow \cos \theta = \frac{\sqrt{2}}{2} \\ 1 = \sqrt{2} \cdot 1 \cdot \sin \theta \Rightarrow \sin \theta = \frac{1}{\sqrt{2}} \end{cases}$$

$$\theta = \frac{\pi}{4}$$

$$\begin{cases} x = 2\rho \cos \theta \\ y = \sqrt{2}\rho \sin \theta \end{cases}$$

$$0 \leq \rho \leq 1$$

$$-\pi/4 \leq \theta \leq \pi/4$$

$$J = 2\sqrt{2}\rho \quad (\text{segundo ejercicio 6) b})$$

$$\iint_D e^{x^2+2y^2} dx dy = \int_0^1 d\rho \int_{-\pi/4}^{\pi/4} d\theta e^{4\rho^2 \cos^2 \theta + 2 \cdot 2\rho^2 \sin^2 \theta} \cdot 2\sqrt{2}\rho =$$

$$2\sqrt{2} \int_0^1 d\rho \int_{-\pi/4}^{\pi/4} d\theta \cdot e^{4\rho^2} \rho = \cancel{2\sqrt{2}} \cdot \frac{\pi}{2} \cdot \int_0^1 e^{4\rho^2} \rho \frac{du}{u/8\rho} =$$

$$\left. \begin{aligned} e^{4\rho^2} &= u \\ 4\rho^2 \cdot 8\rho d\rho &= du \\ \rho=0 &\rightarrow u=e^0=1 \\ \rho=1 &\rightarrow u=e^4 \end{aligned} \right\} = \frac{\sqrt{2}\pi}{8} (e^4 - 1)$$

8)

$$a) \iint_D e^{-x^2-y^2} dx dy = \int_0^{2\pi} d\theta \int_0^\infty dp e^{-p^2} dp p =$$

$$= 2\pi \int_1^0 p \frac{du}{-2up} p = -\pi \int_1^0 du =$$

$$\begin{aligned} e^{-p^2} &= u \\ e^{-p^2} (-2p) dp &= du \end{aligned}$$

$$= (-\pi) (0-1) = \pi$$

$$p=0 \Rightarrow u = e^0 = 1$$

$$p \rightarrow \infty \Rightarrow u = \frac{e}{p \cdot \infty} e^{-p^2} = 0$$

$$8)b) \left(\int_{-\infty}^{\infty} e^{-u^2} du \right)^2 = \left(\int_{-\infty}^{\infty} e^{-u^2} du \right) \cdot \left(\int_{-\infty}^{\infty} e^{-u^2} du \right) = \textcircled{A}$$

$$\begin{aligned} \textcircled{B} &= \iint_D e^{-x^2-y^2} dx dy = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dy e^{-x^2-y^2} = \int_{-\infty}^{+\infty} dx \left[\int_{-\infty}^{+\infty} dy e^{-x^2-y^2} \right] \\ &= \int_{-\infty}^{+\infty} dx \left[\int_{-\infty}^{+\infty} e^{-x^2} \cdot e^{-y^2} dy \right] = \left[\int_{-\infty}^{+\infty} e^{-x^2} dx \right] \cdot \left[\int_{-\infty}^{+\infty} e^{-y^2} dy \right] = \end{aligned}$$

Haciendo en c/caso el cambio de variables

obtenemos

$$\begin{aligned} y &= u \\ x &= u \end{aligned}$$

$$B = \left(\int_{-\infty}^{+\infty} e^{-u^2} du \right) \cdot \left(\int_{-\infty}^{+\infty} e^{-u^2} du \right) = \left[\int_{-\infty}^{+\infty} e^{-u^2} du \right]^2$$

$$8c) \text{ Area} = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{\pi}} e^{-\frac{1}{2} z^2} dz = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2} z^2} dz =$$

Substitución

$$u = -\frac{1}{2} z^2$$

$$du = -\frac{1}{2} dz$$

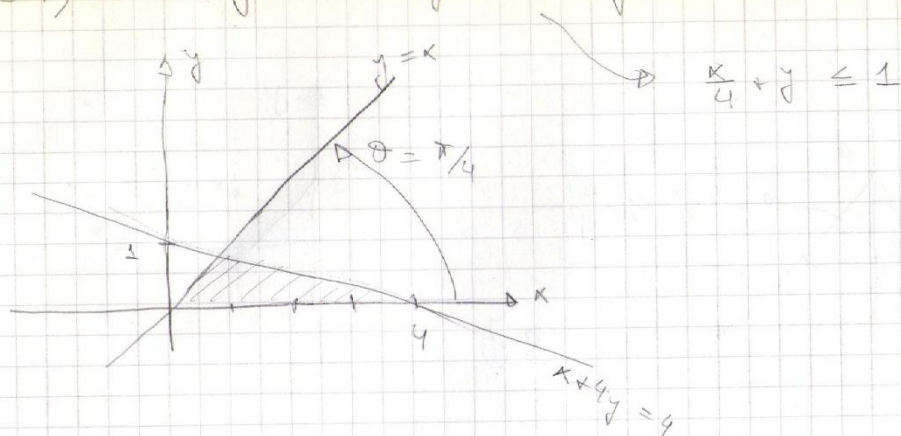
$$z \rightarrow \pm \infty \Rightarrow u \rightarrow -\infty$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-u} \sqrt{2} du =$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-u} du =$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}} \cdot \sqrt{\pi} = 1$$

$$09) \quad x \geq y \quad x+4y \leq 4 \quad y \geq 0$$



$$\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \end{cases}$$

$$x+4y=4 \Rightarrow \rho \cos \theta + 4 \rho \sin \theta = 4 \Rightarrow \rho = \frac{4}{\cos \theta + 4 \sin \theta}$$

$$\iint_D \frac{x+4y}{x^2} dx dy = \int_0^{\pi/4} d\theta \int_0^{\frac{4}{\cos \theta + 4 \sin \theta}} \frac{\rho \cos \theta + 4 \rho \sin \theta}{\rho^2 \cos^2 \theta} \rho d\rho$$

$$= \int_0^{\pi/4} d\theta \left(\frac{\cos \theta + 4 \sin \theta}{\cos^2 \theta} \right) \frac{4}{\cos \theta + 4 \sin \theta} =$$

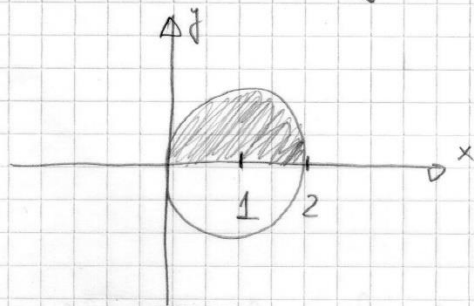
$$= 4 \left[\tan \theta \right]_0^{\pi/4} = 4(1-0) = 4$$

$$10) \quad a) \quad \int_0^{\pi/2} d\varphi \int_0^{2\cos\varphi} p^3 dp \quad \int p \cdot p^2 \quad \int \frac{p^2}{x^2+y^2}$$

$$0 \leq \varphi \leq \pi/2$$

$$0 \leq p \leq 2\cos\varphi$$

$$p = 2\cos\varphi \Rightarrow p^2 = 2p\cos\varphi \Rightarrow x^2 + y^2 = 2x \Rightarrow x^2 - 2x + 1 + y^2 = 1 \Rightarrow (x-1)^2 + y^2 = 1$$



$$\int_0^2 dx \int_0^{\sqrt{2x-x^2}} (x^2+y^2) dy$$

Resuelvo en polares:

$$\int_0^{\pi/2} d\varphi \int_0^{2\cos\varphi} p^3 dp = \int_0^{\pi/2} d\varphi \left[\frac{p^4}{4} \Big|_0^{2\cos\varphi} \right] =$$

$$= \int_0^{\pi/2} d\varphi \left[\frac{16}{4} \cos^4\varphi \right] = 4 \int_0^{\pi/2} \cos^4\varphi d\varphi =$$

$$= 4 \left[\frac{3}{8} \varphi + \frac{\sin(2\varphi)}{4} + \frac{\sin(4\varphi)}{32} \right]_0^{\pi/2} =$$

$$= 4 \left[\frac{3}{8} \frac{\pi}{2} + \frac{\sin \pi}{4} + \frac{\sin(2\pi)}{32} - 0 \right] = \frac{3}{4} \pi$$

$$b) \int_{-\pi/6}^{\pi/3} d\varphi \int_0^{\frac{\sqrt{3}}{\cos \varphi}}$$

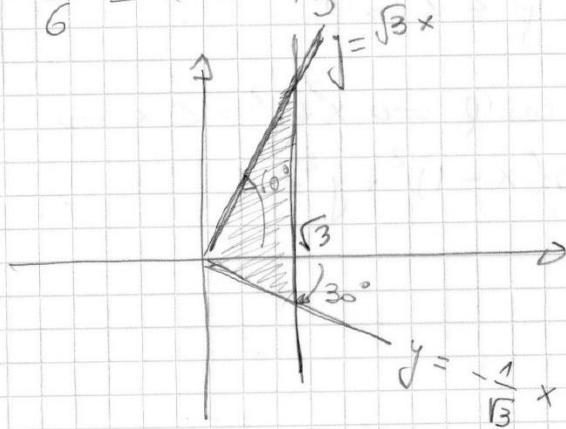
$$\int_0^2 \cos \varphi \, d\rho$$

$$\int_0^{\frac{\sqrt{3}}{\cos \varphi}} \frac{\rho \cos \varphi}{x} \, d\rho$$

$$-\frac{\pi}{6} \leq \varphi \leq \frac{\pi}{3}$$

$$0 \leq \rho \leq \frac{\sqrt{3}}{\cos \varphi}$$

$$\rho \cos \varphi \leq \sqrt{3} \Rightarrow x \leq \sqrt{3}$$



$$\int_0^{\sqrt{3}} dx \int_{-\frac{1}{\sqrt{3}}x}^{\sqrt{3}x} x \, dy$$

Resuelvo en cartesianas

$$\int_0^{\sqrt{3}} dx \, x \left[\sqrt{3}x + \frac{1}{\sqrt{3}}x \right] = \left(\sqrt{3} + \frac{1}{\sqrt{3}} \right) \int_0^{\sqrt{3}} x^2 \, dx =$$

$$= \left(\sqrt{3} + \frac{1}{\sqrt{3}} \right) \left[\frac{x^3}{3} \right]_0^{\sqrt{3}} = \left(\sqrt{3} + \frac{1}{\sqrt{3}} \right) \frac{\sqrt{3}}{3} =$$

$$= \frac{3 + 1}{\sqrt{3}} \cdot \sqrt{3} = 4$$

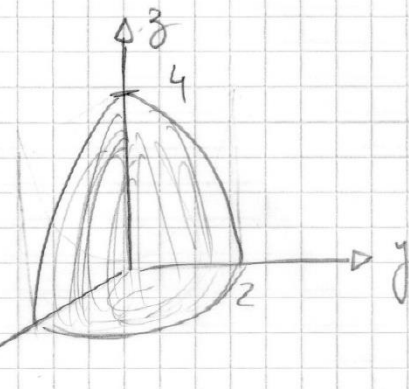
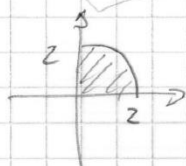
$$11) \int_0^{\pi/2} d\varphi \int_0^2 \rho \, d\rho \int_0^{4-\rho^2} \rho^2 \, dz$$

$\underbrace{\rho \cdot \rho}_{\text{área}} \cdot \underbrace{\rho \, dz}_{\text{altura}} = \rho^2 \, dz$
 $\sqrt{x^2+y^2}$

$$0 \leq \varphi \leq \pi/2$$

$$0 \leq \rho \leq 2$$

$$0 \leq z \leq \sqrt{4-x^2-y^2}$$



$$\int_0^2 dx \int_0^{\sqrt{4-x^2}} dy \int_0^{\sqrt{4-x^2-y^2}} \sqrt{x^2+y^2} \, dz$$

Resuelto en polares:

$$\int_0^{\pi/2} d\varphi \int_0^2 \rho \, d\rho \int_0^{4-\rho^2} \rho^2 \, dz = \int_0^{\pi/2} d\varphi \int_0^2 \underbrace{\rho^2 (4-\rho^2)}_{4\rho^2 - \rho^4} \, d\rho$$

$$= \frac{\pi}{2} \left[4 \frac{\rho^3}{3} - \frac{\rho^5}{5} \right]_0^2 = \frac{\pi}{2} \left[\frac{4 \cdot 8}{3} - \frac{32}{5} \right] =$$

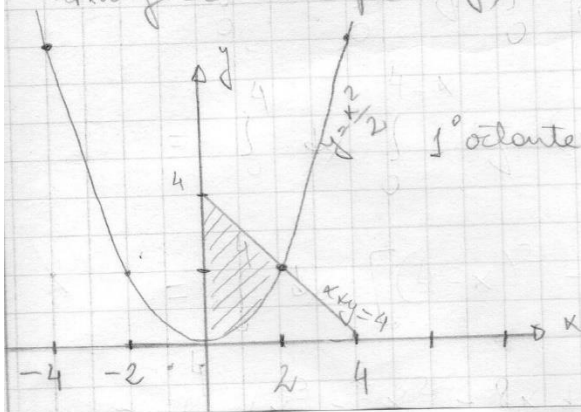
$$= \frac{\pi}{2} \cdot \frac{64}{15} = \frac{32\pi}{15}$$

12) a) $2y \geq x^2 + z$; $x+y \leq 4$; 1^o octante

$$z \leq 2y - x^2$$

si $z=0 \Rightarrow 2y - x^2 \geq 0 \Rightarrow y \geq \frac{x^2}{2}$

Projecc. en el (xy)



$$\begin{cases} y = \frac{x^2}{2} \\ y = 4-x \end{cases} \Rightarrow \frac{x^2}{2} = 4-x \Rightarrow$$

$$\Rightarrow \frac{x^2}{2} + x - 4 = 0 \Rightarrow x^2 + 2x - 8 = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{4+32}}{2} \Rightarrow x = 2 \text{ or } -4$$

$$\text{Vol } V = \int_0^2 dx \int_{\frac{x^2}{2}}^{4-x} dy \int_0^{2y-x^2} dz = \int_0^2 dx \int_{\frac{x^2}{2}}^{4-x} (2y-x^2) dy =$$

$$= \int_0^2 dx \left[y^2 - x^2 y \right]_{\frac{x^2}{2}}^{4-x} = \int_0^2 dx \left[(4-x)^2 - x^2(4-x) - \frac{x^4}{4} + \right.$$

$$\left. + x^2 \frac{x^2}{2} \right] = \int_0^2 (16 + x^2 - 8x - 4x^2 + x^3 + \frac{x^4}{4}) dx =$$

$$= \left[16x - \frac{x^3}{3} - 4x^2 + \frac{x^4}{4} + \frac{x^5}{20} \right]_0^2 = 32 - 8 - 16 + 4 + 1.6 =$$

$$= 13.6$$

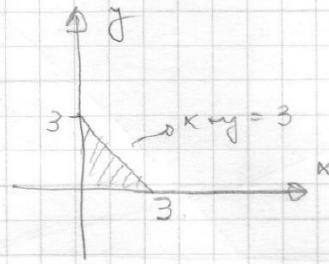
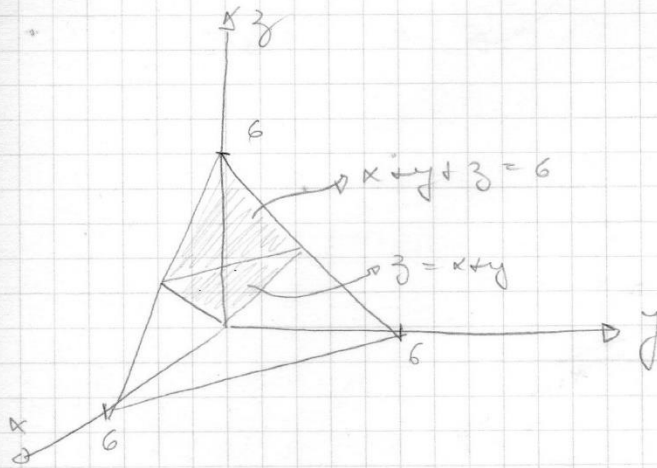
12) b)

$$x + y + z \leq 6$$

$$z \geq x + y$$

$$x \geq 0$$

$$y \geq 0$$



$$\left. \begin{array}{l} x + y + z = 6 \\ z = x + y \end{array} \right\} \Rightarrow 0 \quad x + y + x + y = 6 \Rightarrow 2(x + y) = 6 \Rightarrow x + y = 3$$

$$\text{Vol } V = \int_0^3 dx \int_0^{3-x} dy \int_{x+y}^{6-x-y} dz = \int_0^3 dx \int_0^{3-x} dy [6 - x - y - x - y] =$$

$$= \int_0^3 dx \int_0^{3-x} [6 - 2x - 2y] dy = \int_0^3 dx [6y - 2xy - y^2]_0^{3-x} =$$

$$= \int_0^3 dx [6(3-x) - 2x(3-x) - (3-x)^2] =$$

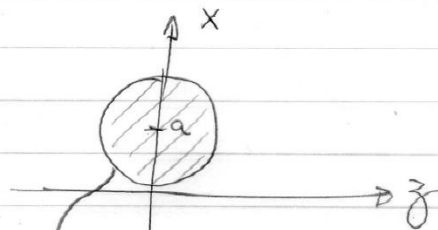
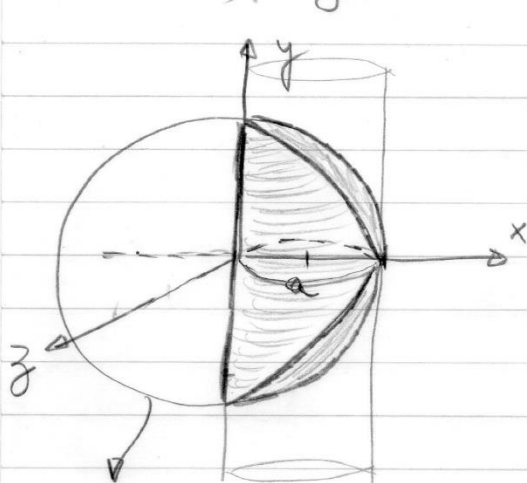
$$= \int_0^3 [18 - 6x - 6x + 2x^2 - 9 - x^2 + 6x] dx =$$

$$= \int_0^3 [x^2 - 6x + 9] dx = \left[\frac{x^3}{3} - 3x^2 + 9x \right]_0^3 =$$

$$= \frac{9}{3} - 27 + 27 = 9$$

12)c) $x^2 + z^2 \leq 2ax$ interior a la esfera de radio $2a$ con centro en $\bar{O}$

$$x^2 + z^2 = 2ax \Rightarrow x^2 - 2ax + z^2 = 0 \Rightarrow x^2 - 2ax + a^2 + z^2 = a^2 \\ \Rightarrow (x-a)^2 + z^2 = a^2$$



$$x^2 + z^2 = 2ax \Rightarrow \\ \Rightarrow \rho^2 = 2a\rho \sin\theta \Rightarrow \\ \Rightarrow \rho = 2a \sin\theta$$

$$x^2 + y^2 + z^2 = 4a^2 \Rightarrow |y| = \sqrt{4a^2 - x^2 - z^2} \Rightarrow |y| = \sqrt{4a^2 - \rho^2}$$

$$\text{Vol } V = \iiint_V dx dy dz = \int_0^\pi d\theta \int_0^{2a \sin\theta} d\rho \int_{-\sqrt{4a^2 - \rho^2}}^{\sqrt{4a^2 - \rho^2}} dy =$$

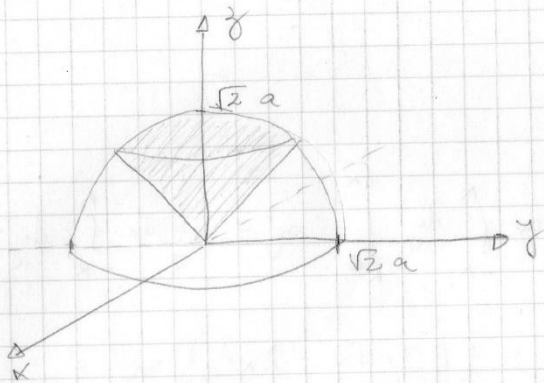
$$= \int_0^\pi d\theta \int_0^{2a \sin\theta} d\rho \left[\rho y \right]_{-\sqrt{4a^2 - \rho^2}}^{\sqrt{4a^2 - \rho^2}} =$$

$$= \int_0^\pi d\theta \int_0^{2a \sin\theta} 2\rho \sqrt{4a^2 - \rho^2} d\rho = \int_0^\pi d\theta \left[-\frac{2}{3} \sqrt{4a^2 - \rho^2}^3 \right]_0^{2a \sin\theta} =$$

$$= \int_0^\pi d\theta \left[-\frac{2}{3} \sqrt{4a^2 - 4a^2 \sin^2\theta}^3 + \frac{2}{3} \sqrt{4a^2}^3 \right] =$$

$$= \frac{16}{9} a^3 (3\pi - 4)$$

$$d) \quad z \geq \sqrt{x^2 + y^2}, \quad x^2 + y^2 + z^2 \leq 2a^2$$



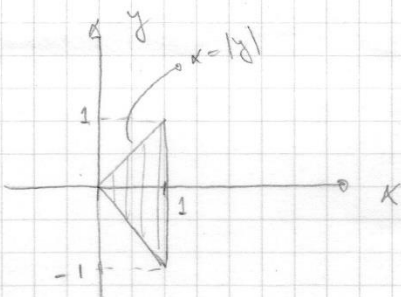
$$\text{Vol} = \iiint_V dx dy dz = \int_0^{2\pi} d\sigma \int_0^{\pi/4} d\lambda \int_0^{\sqrt{2}a} dr \cdot r^2 \sin\lambda =$$

$$= 2\pi \int_0^{\pi/4} \sin\lambda d\lambda \left[\frac{r^3}{3} \right]_0^{\sqrt{2}a} = 2\pi \frac{2\sqrt{2}a^3}{3} \left[-\cos\lambda \right]_0^{\pi/4} =$$

$$= \frac{4\pi\sqrt{2}a^3}{3} \left(-\frac{\sqrt{2}}{2} + 1 \right) = \frac{4\pi\sqrt{2}a^3}{3} \left(1 - \frac{1}{\sqrt{2}} \right) = (\sqrt{2}-1) \frac{4\pi a^3}{3}$$

$$e) \quad z \geq x^2, \quad x \geq z^2 \rightarrow |z| \leq \sqrt{x}$$

$$\left. \begin{array}{l} z = x^2 \\ x = z^2 \end{array} \right\} \Rightarrow z = z^2 = 0 \quad z(z-1) = 0 \Rightarrow \underbrace{z=0 \vee z=1}_{\text{luego } z \geq 0, \text{ entonces es } z \leq \sqrt{x}}$$

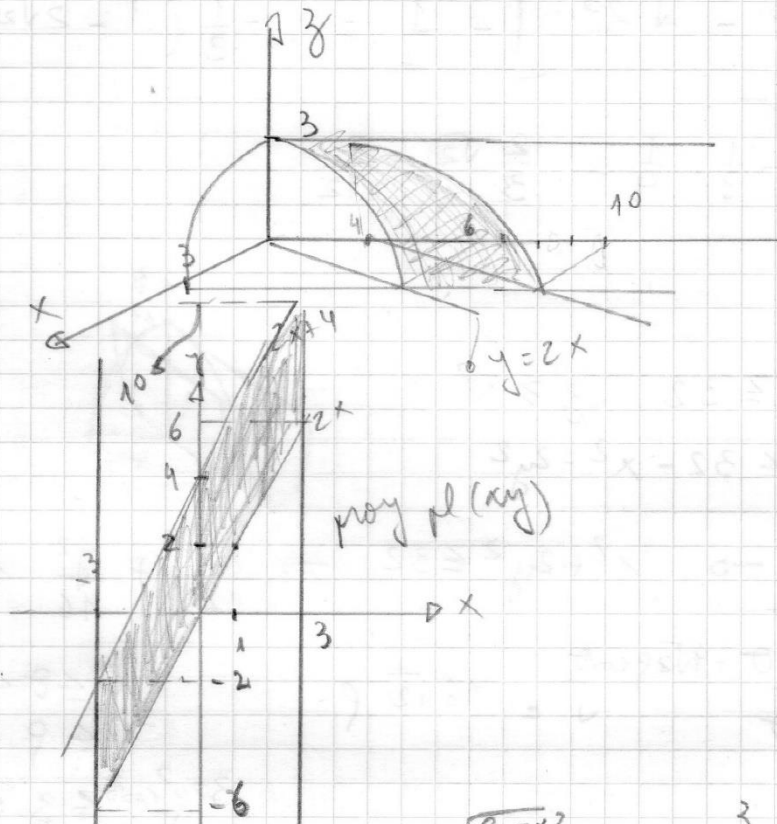


$$\text{Vol } V = \iiint_V dx \, dy \, dz = \int_0^1 dx \int_{-x}^x dy \int_{x^2}^{\sqrt{x}} dz = \int_0^1 dx \int_{-x}^x dy (\sqrt{x} - x^2)$$

$$= \int_0^1 dx (x+x) (\sqrt{x} - x^2) = \int_0^1 (2 \underbrace{x \sqrt{x}}_{x^{3/2}} - 2x^3) dx =$$

$$= \left[2 \cdot \frac{2}{5} x^{5/2} - 2 \frac{x^4}{4} \right]_0^1 = \frac{4}{5} - \frac{1}{2} = \frac{8-5}{10} = \frac{3}{10}$$

12) f) $x^2 + z^2 \leq 9$, $y \geq 2x$, $y \leq 2x+4$



$$\text{Vol } H = \int_{-3}^3 dx \int_{2x}^{2x+4} dy \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dz = \int_{-3}^3 dx (2x+4-2x) \cdot 2\sqrt{9-x^2}$$

$$= \int_{-3}^3 4 \cdot 2\sqrt{9-x^2} dx = 8 \left[\frac{1}{2} \left(x\sqrt{9-x^2} + \right.$$

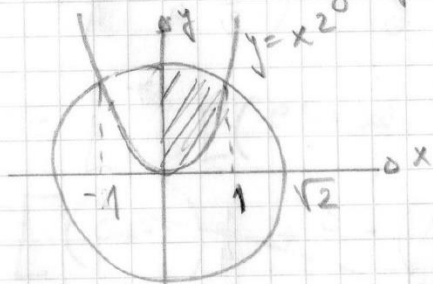
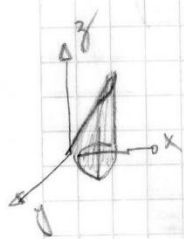
$$\left. + 9 \arccos \frac{x}{3} \right) \Big|_{-3}^3 = \frac{8}{2} \left[3 \cdot 0 + 9 \arccos 1 - (-3) \cdot 0 - 9 \arccos(-1) \right]$$

$$= 4 \cdot 9 \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = 36 \frac{\pi}{2} \cdot 2 = 36\pi$$

g)

$$y \geq x^2 \quad x^2 + y^2 \leq 2 \quad \begin{matrix} z \geq 0 \\ z \leq x \end{matrix} \rightarrow x \geq 0$$

grafico sólo proyección



$$\begin{cases} y = x^2 \\ x^2 + y^2 = 2 \end{cases}$$

$$\Rightarrow y + y^2 = 2 \Rightarrow$$

$$\Rightarrow y^2 + y - 2 = 0 \quad \begin{matrix} y = 1 \rightarrow x^2 = 1 \rightarrow x = 1 \\ y = -2 \rightarrow x^2 = -2 \text{ no} \end{matrix}$$

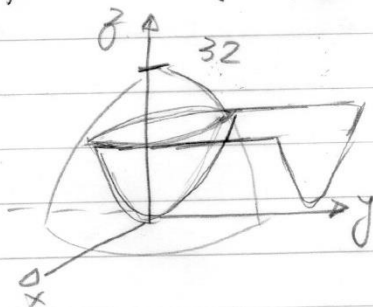
$$\text{Vol H} = \int_0^1 dx \int_{x^2}^{\sqrt{2-x^2}} dy \int_0^x dz = \int_0^1 (\sqrt{2-x^2} - x^2) x dx = \int_0^1 x \sqrt{2-x^2} dx -$$

$$- \int_0^1 x^3 dx = -\frac{1}{3} \left[\sqrt{(2-x^2)^3} \right]_0^1 - \left[\frac{x^4}{4} \right]_0^1 =$$

$$= -\frac{1}{3} \left[\sqrt{(2-1)^3} - \sqrt{2^3} \right] - \frac{1}{4} = -\frac{1}{3} (1 - 2\sqrt{2}) - \frac{1}{4} =$$

$$= \frac{2}{3} \sqrt{2} - \frac{1}{3} - \frac{1}{4} = \frac{2}{3} \sqrt{2} - \frac{7}{12}$$

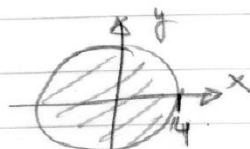
$$h) \quad x^2 + 2y^2 + z \leq 32 \wedge z \geq x^2$$



Analizo la proyección de la curva de corte en el pl(xy)

$$\begin{cases} z = 32 - x^2 - 2y^2 \\ z = x^2 \end{cases} \Rightarrow$$

$$\Rightarrow 32 - x^2 - 2y^2 = x^2 \Rightarrow x^2 + y^2 = 16$$



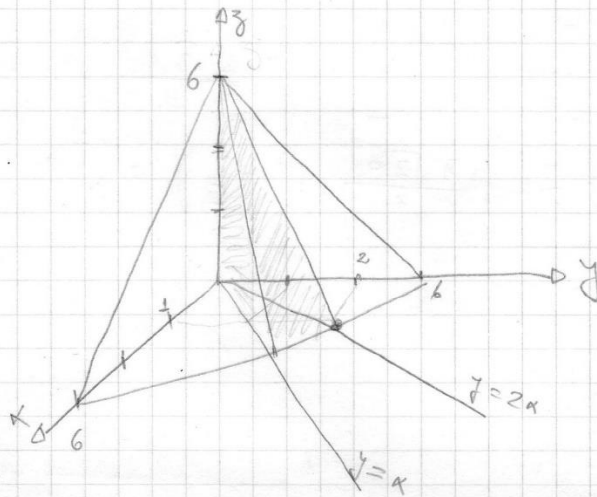
$$\text{Vol H} = \iiint_V dx dy dz = \int_0^{2\pi} d\theta \int_0^4 dp \int_{p^2 \cos^2 \theta}^{32 - p^2 \cos^2 \theta - 2p^2 \sin^2 \theta} p dz =$$

$$= \int_0^{2\pi} d\theta \int_0^4 dp \, p [32 - p^2 \cos^2 \theta - 2p^2 \sin^2 \theta - p^2 \cos^2 \theta] =$$

$$= \int_0^{2\pi} d\theta \int_0^4 dp \, p [32 - 2p^2] = \int_0^{2\pi} d\theta \left[\frac{32p^2}{2} - \frac{2p^4}{4} \right]_0^4 =$$

$$= \int_0^{2\pi} d\theta \left(16 \cdot 16 - \frac{1}{2} 256 \right) = 2\pi \cdot 128 = \boxed{256\pi}$$

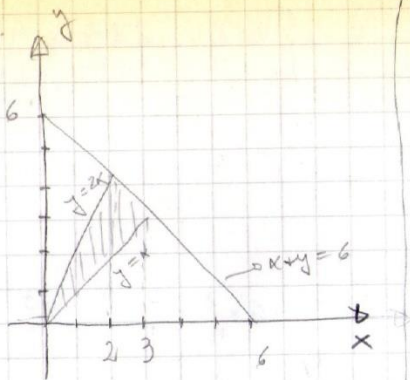
13)



$$\delta = k \cdot \text{dist} \text{ to } (yz)$$

$$\delta = k|x| \quad (k \in \mathbb{R})$$

$$\delta = kx \quad \text{for the 1st octant}$$



$$\begin{aligned} \text{Mass} &= \iiint \delta \, dx \, dy \, dz = \iiint kx \, dx \, dy \, dz = \\ &= \int_0^2 dx \int_x^{2x} dy \int_0^{6-x-y} kx \, dz + \int_2^3 dx \int_x^{6-x} dy \int_0^{6-x-y} kx \, dz \\ &= \frac{15}{2} \end{aligned}$$

$$\left. \begin{array}{l} y=x \\ x+y=6 \end{array} \right\} \Rightarrow 2x=6 \Rightarrow x=3$$

$$\left. \begin{array}{l} y=2x \\ x+y=6 \end{array} \right\} \Rightarrow x+2x=6 \Rightarrow x=\frac{6}{3}=2$$

14)

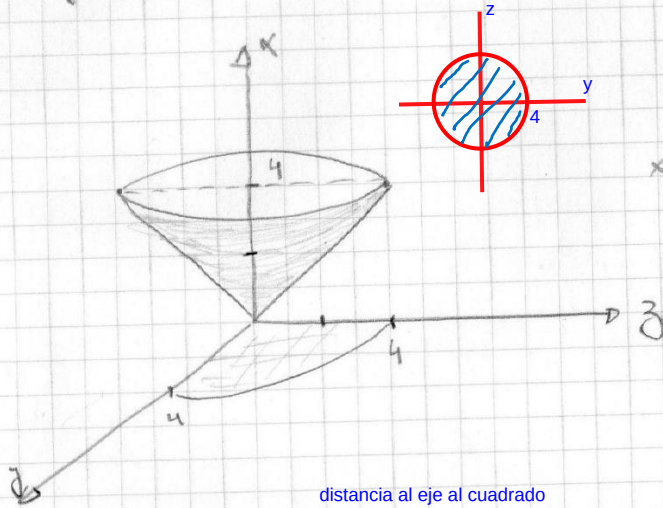
$$\sqrt{y^2 + z^2} \leq x$$

frontera

$$x = \sqrt{y^2 + z^2} \Rightarrow x^2 = y^2 + z^2$$

$$x = 4 \Rightarrow y^2 + z^2 = 16$$

$$x = \sqrt{\underbrace{\rho^2 \sin^2 \theta}_{z^2} + \underbrace{\rho^2 \cos^2 \theta}_{y^2}} = \sqrt{\rho^2} = \rho$$



distancia al eje al cuadrado

$$I_x = \iiint_V \delta(y^2 + z^2) dx dy dz = \iiint_V k |y| (y^2 + z^2) dx dy dz$$

$$= k \int_0^{2\pi} d\theta \int_0^4 \rho d\rho \int_\rho^4 dx |\rho \cos \theta| \rho^2 \rho =$$

$$= k \int_0^{2\pi} d\theta \int_0^4 \rho d\rho \int_\rho^4 \rho^4 |\cos \theta| dx =$$

$$= k \int_0^{2\pi} d\theta \int_0^4 dp (4-p) p^4 |\cos \theta| = k \int_0^{2\pi} |\cos \theta| d\theta \int_0^4 (4p^4 - p^5) dp$$

$$= k \int_0^{2\pi} |\cos \theta| d\theta \left[\frac{4p^5}{5} - \frac{p^6}{6} \right]_0^4 = k \left(\frac{4096}{5} - \frac{4096}{6} \right).$$

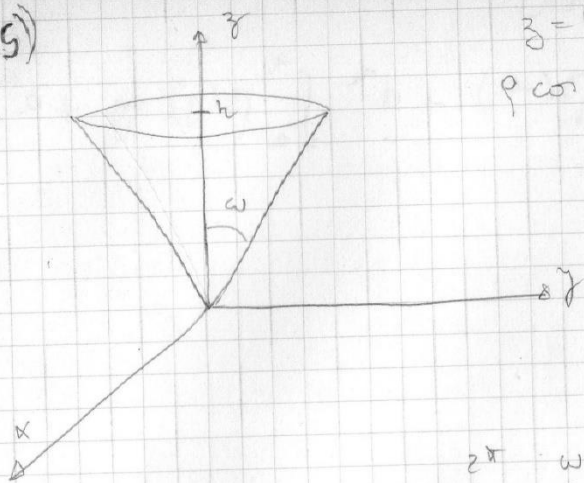
$$= \left[\int_0^{\pi/2} \cos \theta d\theta + \int_{\pi/2}^{3\pi/2} -\cos \theta d\theta + \int_{3\pi/2}^{2\pi} \cos \theta d\theta \right] =$$

$$= k \cdot \frac{2048}{15} \left[\sin \theta \Big|_0^{\pi/2} - \sin \theta \Big|_{\pi/2}^{3\pi/2} + \sin \theta \Big|_{3\pi/2}^{2\pi} \right] =$$

$$= k \frac{2048}{15} \left[\underset{1}{(1-0)} - \underset{2}{(-1-1)} + \underset{+1}{(0+1)} \right] = k \frac{2048}{15} \cdot 4 =$$

$$= \frac{8192k}{15}$$

15)



$$z = h$$

$$\rho \cos \lambda = h \Rightarrow \rho = \frac{h}{\cos \lambda}$$



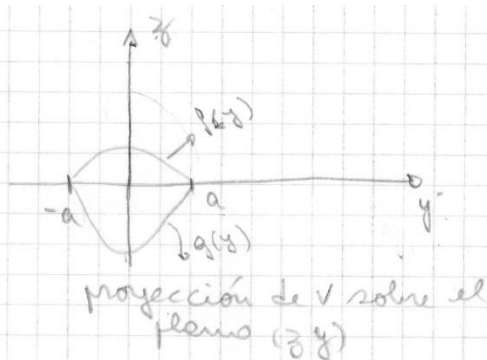
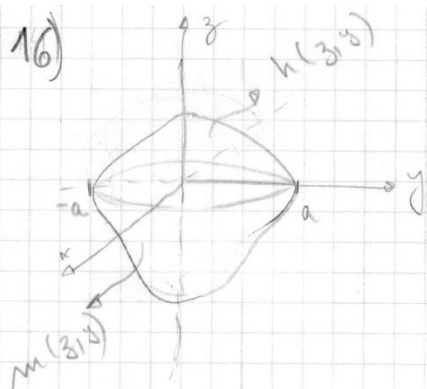
$$\text{Vol } V = \iiint_V dx dy dz = \int_0^{2\pi} d\varphi \int_0^{\omega} d\lambda \int_0^{\frac{h}{\cos \lambda}} d\rho \rho^2 \sin \lambda =$$

$$= 2\pi \int_0^{\omega} d\lambda \sin \lambda \left[\frac{\rho^3}{3} \right]_0^{\frac{h}{\cos \lambda}} = \frac{2\pi}{3} \int_0^{\omega} \sin \lambda \frac{h^3}{\cos^3 \lambda} d\lambda =$$

$$= \frac{2\pi h^3}{3} \left[\frac{1}{2} \frac{1}{\cos^2 \lambda} \right]_0^{\omega} = \frac{2\pi h^3}{3} \left[\frac{1}{2} \frac{1}{\cos^2 \omega} - \frac{1}{2} \right] =$$

$$= \frac{\pi h^3}{3} \left[\frac{1}{\cos^2 \omega} - 1 \right] = \frac{\pi h^3}{3} \frac{1 - \cos^2 \omega}{\cos^2 \omega} = \frac{\pi h^3}{3} \frac{\sin^2 \omega}{\cos^2 \omega}$$

16)



$$\iiint_V y^n dx dy dz = \int_{-a}^a dy \int_{g(y)}^{f(y)} dz \int_{h(z,y)}^{m(z,y)} dx y^n =$$

$$= \int_{-a}^a dy y^n \int_{g(y)}^{f(y)} dz [m(z,y) - h(z,y)]$$

esta integral es una función de y
sea $\beta(y)$

$$= \int_{-a}^a y^n \beta(y) dy = \frac{1}{2} \left(\frac{y^{n+1}}{n+1} \beta(y) + \int \beta(y) dy \cdot y^n \right) \Big|_{-a}^a$$

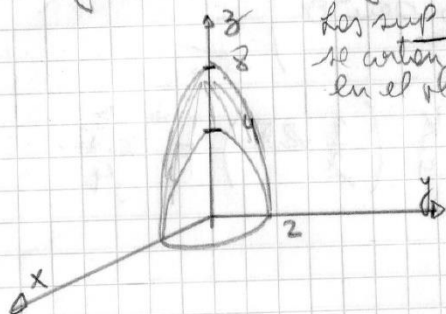
$$= \frac{1}{2} \left(\frac{a^{n+1}}{n+1} \beta(a) + \int_{-a}^a \beta(y) dy \cdot a^n - \frac{(-a)^{n+1}}{n+1} \beta(-a) - (-a)^n \int_{-a}^a \beta(y) dy \right)$$

n impar $\Rightarrow n+1$ es par

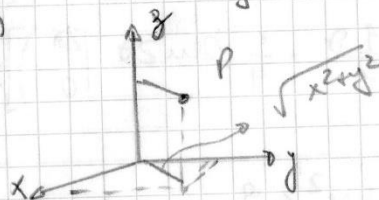
$$= \frac{1}{2} \left(\frac{a^{n+1}}{n+1} \beta(a) - \frac{a^{n+1}}{n+1} \beta(-a) + a^n \int_{-a}^a \beta(y) dy + a^n \int_{-a}^a \beta(y) dy \right)$$

17) a) $z = 4 - x^2 - y^2$ $z = 8 - 2x^2 - 2y^2$
 Busco la proyección en el $pl(xy)$ de la curva intersección de las superficies

$$4 - x^2 - y^2 = 8 - 2x^2 - 2y^2 \Rightarrow x^2 + y^2 = 4$$



las sup se cortan en el $pl(xy)$
 $z = 4 - \sqrt{x^2 + y^2} = 0$



$$\delta = k \sqrt{x^2 + y^2} \quad (k \in \mathbb{R}^+)$$

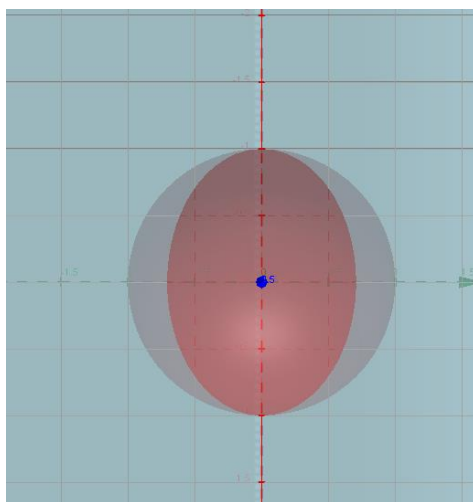
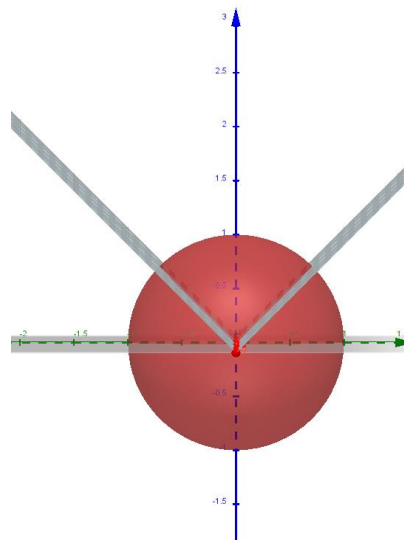
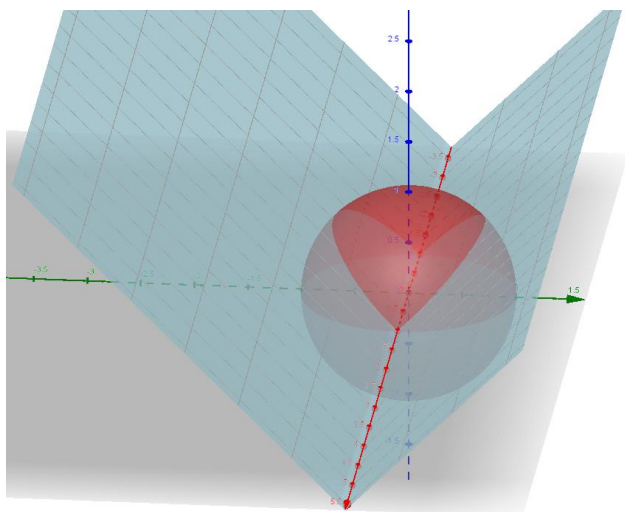
$$M = \iiint_V \delta \, dx \, dy \, dz = k \int_0^{2\pi} d\theta \int_0^2 dp \int_{\frac{4-p^2}{2}}^{8-2p^2} dz \, p \sqrt{p^2} =$$

$$= k \, 2\pi \int_0^2 dp \, p^2 (8 - 2p^2 - 4 + p^2) = 2k\pi \int_0^2 dp (4p^2 - p^4) =$$

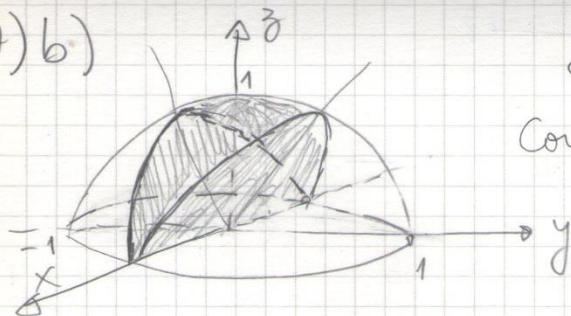
$$= 2k\pi \left(\frac{4p^3}{3} - \frac{p^5}{5} \right) \Big|_0^2 = 2k\pi \left(\frac{32}{3} - \frac{32}{5} \right) = 2k\pi \frac{64}{15} = \frac{128k\pi}{15}$$

17)b)

Calcule la masa del cuerpo definido por: $z \geq \text{abs}(y)$, $x^2 + y^2 + z^2 \leq 1$ si la densidad en cada punto es proporcional a la distancia desde el punto al plano xy



17) b)



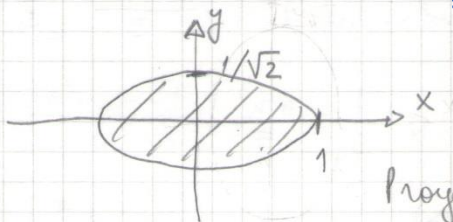
$$\delta = k|z| \quad k \in \mathbb{R}^+$$

Como $0 \leq |y| \leq z \Rightarrow$
 $\Rightarrow z \in \mathbb{R}_0^+$

$$\delta = kz$$

$$\begin{cases} z = |y| \\ x^2 + y^2 + z^2 = 1 \end{cases} \Rightarrow x^2 + y^2 + |y|^2 = 1 \Rightarrow x^2 + 2y^2 = 1$$

$$x^2 + \frac{y^2}{\frac{1}{2}} = 1$$



Proyección en (xy)

$$\begin{cases} x = \rho \cos \theta \\ \sqrt{2}y = \rho \sin \theta \end{cases} \Rightarrow \begin{cases} x = \rho \cos \theta \\ y = \frac{\rho}{\sqrt{2}} \sin \theta \end{cases} \quad \begin{matrix} 0 \leq \theta \leq 2\pi \\ 0 \leq \rho \leq 1 \end{matrix}$$

$$J = 1 \cdot \frac{1}{\sqrt{2}} \cdot \rho$$

$$M = \iiint_V k|z| \, dx \, dy \, dz = k \int_0^{2\pi} d\theta \int_0^1 d\rho \int dz \sqrt{1 - \rho^2 \cos^2 \theta - \frac{\rho^2}{2} \sin^2 \theta} \cdot \frac{\rho}{\sqrt{2}} =$$

$$\frac{k}{\sqrt{2}} \left[\int_0^{\pi} d\theta \int_0^1 d\rho \, \rho \left[\frac{z^2}{2} \sqrt{1 - \rho^2 \cos^2 \theta - \frac{\rho^2}{2} \sin^2 \theta} \right] \right] +$$

$$+ \int_{\pi}^{2\pi} d\theta \int_0^1 d\rho \, \rho \left[\frac{z^2}{2} \sqrt{1 - \rho^2 \cos^2 \theta - \frac{\rho^2}{2} \sin^2 \theta} \right] =$$

$$= \frac{k}{2\sqrt{2}} \left[\int_0^{\pi} d\theta \int_0^1 d\rho \, \rho \left[1 - \rho^2 \cos^2 \theta - \frac{\rho^2}{2} \sin^2 \theta - \frac{\rho^2}{2} \sin^2 \theta \right] + \right.$$

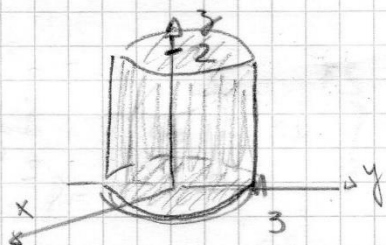
$$\left. + \int_{\pi}^{2\pi} d\theta \int_0^1 d\rho \, \rho \left[1 - \rho^2 \cos^2 \theta - \frac{\rho^2}{2} \sin^2 \theta - \frac{\rho^2}{2} \sin^2 \theta \right] \right] =$$

$$= \frac{k}{2\sqrt{2}} \left[\int_0^{\pi} d\phi \int_0^1 dr \, r(1-r^2) + \int_{\pi}^{2\pi} d\phi \int_0^1 dr \, r(1-r^2) \right] =$$

$$= \frac{k}{2\sqrt{2}} \left[\int_0^{2\pi} d\phi \int_0^1 dr \, (r-r^3) \right] = \frac{k}{2\sqrt{2}} \cdot 2\pi \cdot \left[\frac{r^2}{2} - \frac{r^4}{4} \right]_0^1 =$$

$$= \frac{k\pi}{\sqrt{2}} \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{k\pi}{\sqrt{2}} \cdot \frac{1}{4} = \frac{k\pi}{4\sqrt{2}} = \frac{k\pi \cdot \sqrt{2}}{4 \cdot 2} = \frac{k\pi\sqrt{2}}{8}$$

c) ¹⁷⁾ $x^2 + y^2 \leq 9$ $0 \leq z \leq 2$



$$\delta = k \cdot |y| \quad k \in \mathbb{R}^+$$

$$M = \iiint_V \delta \, dx \, dy \, dz = k \int_0^{2\pi} d\sigma \int_0^3 dp \, p \int_0^2 dz |p \sin \sigma| =$$

$$= k \cdot 2 \int_0^{2\pi} |\sin \sigma| \, d\sigma \int_0^3 dp \, p^2 = 2k \cdot \left[\frac{p^3}{3} \Big|_0^3 \right] \cdot$$

$$\left(\int_0^{\pi} \sin \sigma \, d\sigma + \int_{\pi}^{2\pi} -\sin \sigma \, d\sigma \right) = 2k \cdot 9 \left[-\cos \sigma \Big|_0^{\pi} + \right.$$

$$\left. + \cos \sigma \Big|_{\pi}^{2\pi} \right] = 18k \left[-(-1) + 1 + 1 - (-1) \right] =$$

$$= 18k \cdot 4 = 72k$$

$$18) \text{ Vol } V = \iiint_V dx \, dy \, dz = \int_0^{2\pi} d\theta \int_0^2 dp \int_{8p^2}^{16} dz =$$

$$= \int_0^{2\pi} d\theta \int_0^2 dp \, p (16 - 8p^2) = 2\pi \cdot \left[16 \frac{p^2}{2} - \frac{8p^4}{4} + \frac{p^6}{6} \right]_0^2 = 2\pi \left[8 \cdot 4 - 2 \cdot 16 + \frac{64}{6} \right]$$

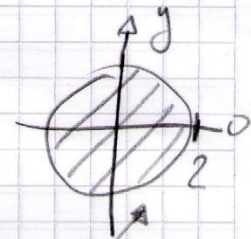
$$= 2\pi \cdot \frac{32}{3} = \left(\frac{64}{3} \pi \right)$$

Busco el máximo de la función

$$f(x,y) = 8(x^2+y^2) - (x^2+y^2)^2$$

$$\left\{ \begin{array}{l} f'_x = 8 \cdot 2x - 2(x^2+y^2) \cdot 2x = 0 \quad (1) \\ f'_y = 8 \cdot 2y - 2(x^2+y^2) \cdot 2y = 0 \quad (2) \end{array} \right.$$

$$\left\{ \begin{array}{l} f'_x = 8 \cdot 2x - 2(x^2+y^2) \cdot 2x = 0 \quad (1) \\ f'_y = 8 \cdot 2y - 2(x^2+y^2) \cdot 2y = 0 \quad (2) \end{array} \right.$$



$$(1) \rightarrow 4x(4 - x^2 - y^2) = 0 \rightarrow x=0 \rightarrow \text{descarto (es mín)}$$

$$(2) \rightarrow 4y(4 - x^2 - y^2) = 0 \rightarrow y=0 \rightarrow \text{descarto (es mín)}$$

$$z_{\text{máx}} = 8 \cdot \underbrace{4}_{x^2+y^2} - \underbrace{4^2}_{x^2+y^2} = 32 - 16 = 16$$